Road Anomaly Detection

A Comparative Study of Classification, Semantic Segmentation and Anomaly Detection

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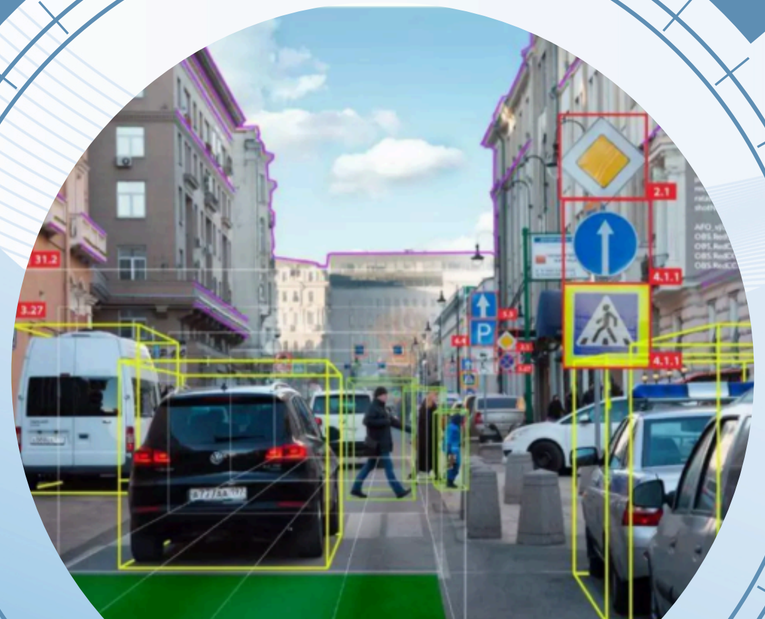
Presentation Roadmap



Problem & objectives

Problem statement and project objectives (BO-DSO)

Data & preprocessing

Dataset exploration and preprocessing.

Models & methodology

Implemented models and experimental methodology.

Results & Interpretation

Results, performance comparisons, and interpretation

Conclusions & next steps

Main conclusions, limitations, and possible improvements.

PROBLEM and OBJECTIVES

THE CHALLENGE OF UNDERSTANDING ROAD SCENES

Animals crossing or standing on roads

Obstacles and abandoned objects

Traffic cones and road works

Unexpected vehicles or objects

1 · Detect
Is something abnormal on the road?

→

2 · Classify
Animal, obstacle or other (DSO 2)

→

3 · Localize
Pixel-level mask of the anomaly (DSO 1)

How can we automatically detect road anomalies, identify their categories and locate them at the pixel level? BO: safer autonomous driving.

FROM BUSINESS NEEDS TO DATA SCIENCE OBJECTIVES

Business objective (BO)

- Improve road scene understanding
- Identify potentially dangerous situations
- Support intelligent driving decisions
- Contribute to road safety

→

Data science objective (DSO)

- Image classification
- Pixel-level anomaly segmentation
- Compare deep learning architectures
- Evaluate detection reliability

Business need → **Intelligent road perception** → **Classification + segmentation** → **Anomaly localization**

DATA and PREPROCESSING

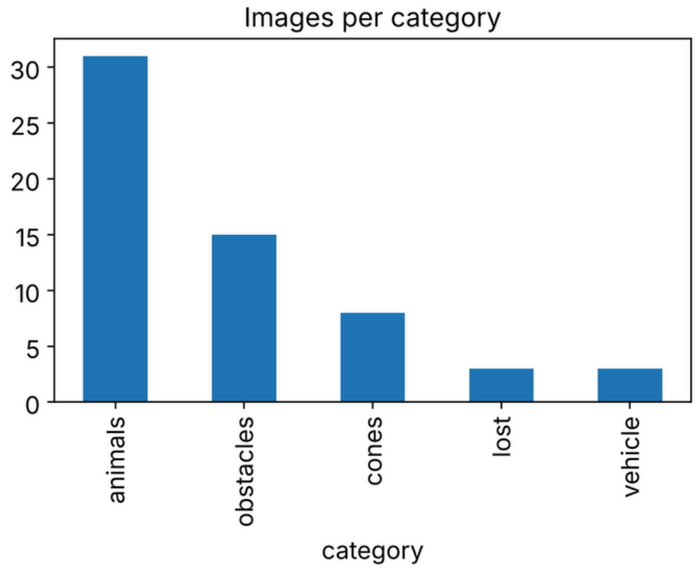
UNDERSTANDING THE ROADANOMALY DATASET

60
Total images

60
Ground truth masks

1280×720
Original resolution

9.85%
Average anomaly area



Category	Images	Share
Animals	**31**	**51.7%**
Obstacles	15	25.0%
Cones	8	13.3%
Lost tires	3	5.0%
Vehicles	3	5.0%

Key observation

Highly imbalanced dataset: animals represent about 51.7% of all images.

FROM RAW IMAGES TO GROUND TRUTH

Understanding pixel-level annotations

01

ORIGINAL IMAGE

Raw road scene

02

GROUND TRUTH

Binary reference mask

03

OVERLAY

Highlighted anomaly

→ ANNOTATION → BINARIZATION → VISUALIZATION →

PIXEL ANNOTATION ENCODING

Original mask

- 0 — Normal background
- 2 — Anomaly

Binary mask

- 0 — Normal
- 1 — Anomaly

GROUND TRUTH = REFERENCE FOR MODEL EVALUATION

DATA PREPARATION PIPELINE

Preparing data for reliable training

1 STAGE 1 INPUT DATA

60
images

1280 × 720
pixels

2 STAGE 2 PREPROCESSING

- Resizing: 512 × 288
- ImageNet normalization

3 STAGE 3 DATA AUGMENTATION

- Horizontal flip
- Rotation
- Scaling
- Brightness
- Color transformations

4 STAGE 4 DATA SPLITTING

TRAIN	34 images	56.7%
VALIDATION	14 images	23.3%
TEST	12 images	20%

TECHNICAL DETAIL

StratifiedGroupKFold — group-aware splitting to reduce data leakage

MODELS and METHODOLOGY

ONE DATASET, TWO DEEP LEARNING TASKS

A PIPELINE A
IMAGE CLASSIFICATION

Road image

→ Small CNN / ResNet34
Classifier

→ PREDICTED CLASS
Animals | Obstacles | Other
one class per image

QUESTION
WHAT
is the anomaly?

B PIPELINE B
SEMANTIC SEGMENTATION

Road image

→ U-Net / DeepLabV3+
Segmenter

→ PIXEL-LEVEL MASK

Normal pixels
Anomalous pixels

QUESTION
WHERE
is the anomaly?

CLASSIFICATION
= CATEGORY RECOGNITION

SEGMENTATION
= SPATIAL LOCALIZATION

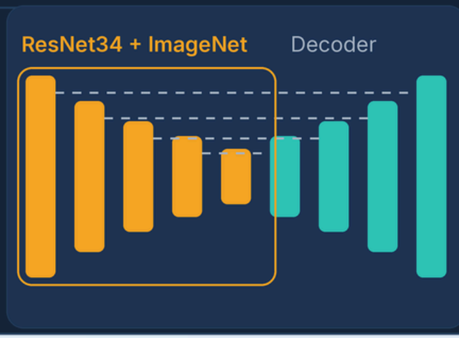
FOUR SEGMENTATION ARCHITECTURES

01 U-NET FROM SCRATCH

- Encoder–decoder
- Skip connections
- No pretrained weights
- Baseline model

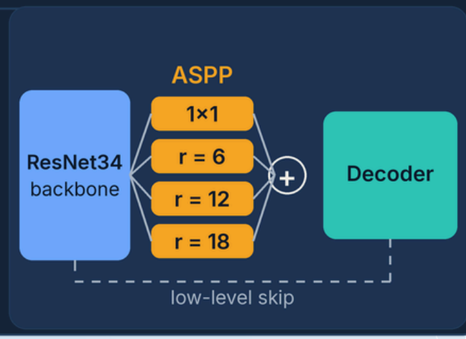
02 U-NET + RESNET34

- ResNet34 encoder
- ImageNet pretrained weights
- Transfer Learning



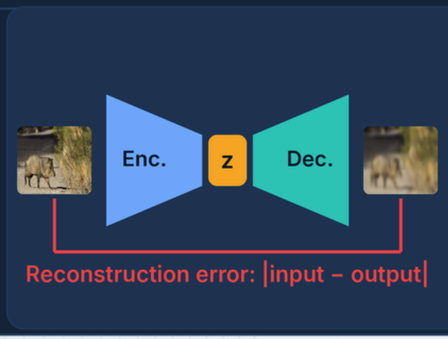
03 DEEPLABV3+ + RESNET34

- ResNet34 backbone
- Atrous Spatial Pyramid Pooling (ASPP)
- Multi-scale contextual information



04 CONVOLUTIONAL AUTOENCODER

- Encoder–decoder reconstruction
- Reconstruction error
- Unsupervised anomaly detection approach



SUPERVISED SEGMENTATION vs RECONSTRUCTION-BASED ANOMALY DETECTION

TRAINING CONFIGURATION and OPTIMIZATION

FRAMEWORK PyTorch	ENVIRONMENT Google Colab	GPU NVIDIA T4	OPTIMIZER AdamW
WEIGHT DECAY 0.0001	BATCH SIZE 4 (segmentation)	LOSS FUNCTION BCE + Dice Loss	SCHEDULER CosineAnnealingLR

TRAINING EPOCHS

U-Net Scratch — 60

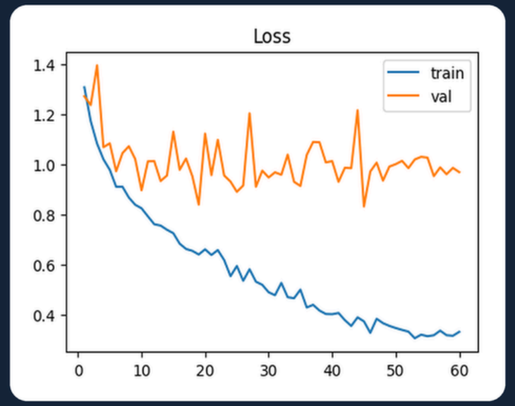
U-Net + ResNet34 — 40

DeepLabV3+ — 40

OPTIMIZATION STRATEGY

- Best checkpoint selected using validation Dice.
- Lower learning rate for pretrained encoder.
- Higher learning rate for newly initialized layers.
- Segmentation threshold: 0.5.

TRAINING LOSS

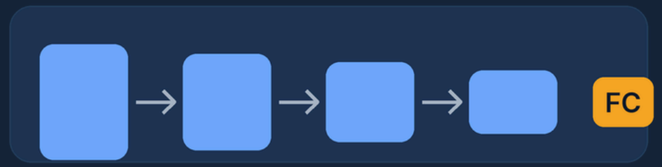


U-Net from scratch · train vs validation loss

IMAGE CLASSIFICATION ARCHITECTURES

A MODEL A
SMALL CNN

- Trained from scratch
- Learns features from our dataset
- Simple baseline



COMPARISON

INITIALIZATION

Random weights VS ImageNet weights

FEATURES

Learned from our data VS Transferred knowledge

ROLE

Simple baseline VS Transfer learning

B MODEL B
RESNET34

- ImageNet pretrained
- Residual connections
- Transfer Learning

CLASSIFICATION OUTPUT

ANIMALS
31 images

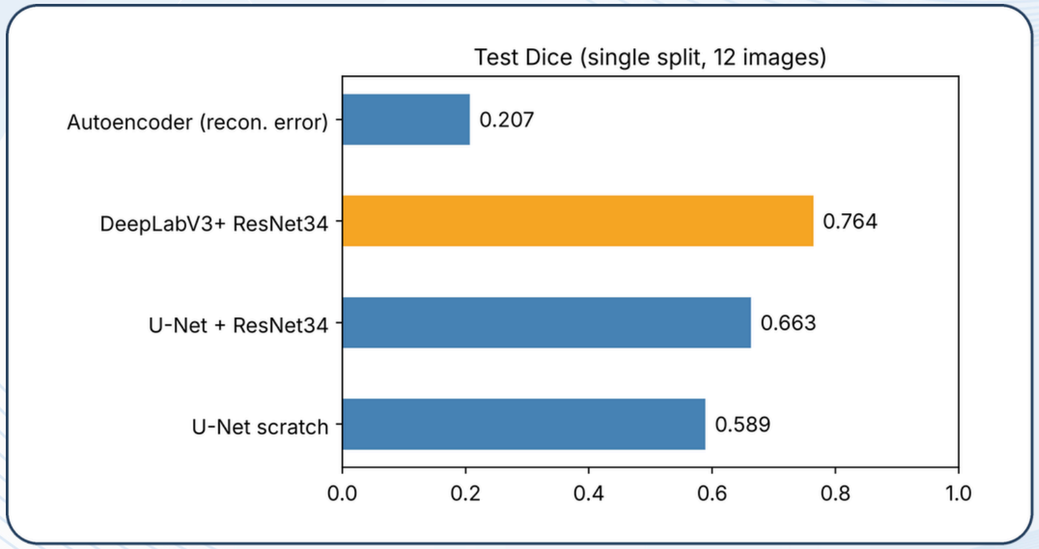
OBSTACLES
15 images

OTHER
14 images

Cones + Lost tires + Vehicles

RESULTS and INTERPRETATION

SEGMENTATION PERFORMANCE COMPARISON



CONCRETE RESULTS

Model	Precision	Recall	Dice	IoU
U-Net Scratch	0.671	0.524	0.589	0.417
U-Net + ResNet34	0.708	0.623	0.663	0.496
DeepLabV3+	**0.930**	**0.648**	**0.764**	**0.618**
Autoencoder	0.121	0.698	0.207	0.115

BEST MODEL

DEEPLABV3+

76.4% DICE

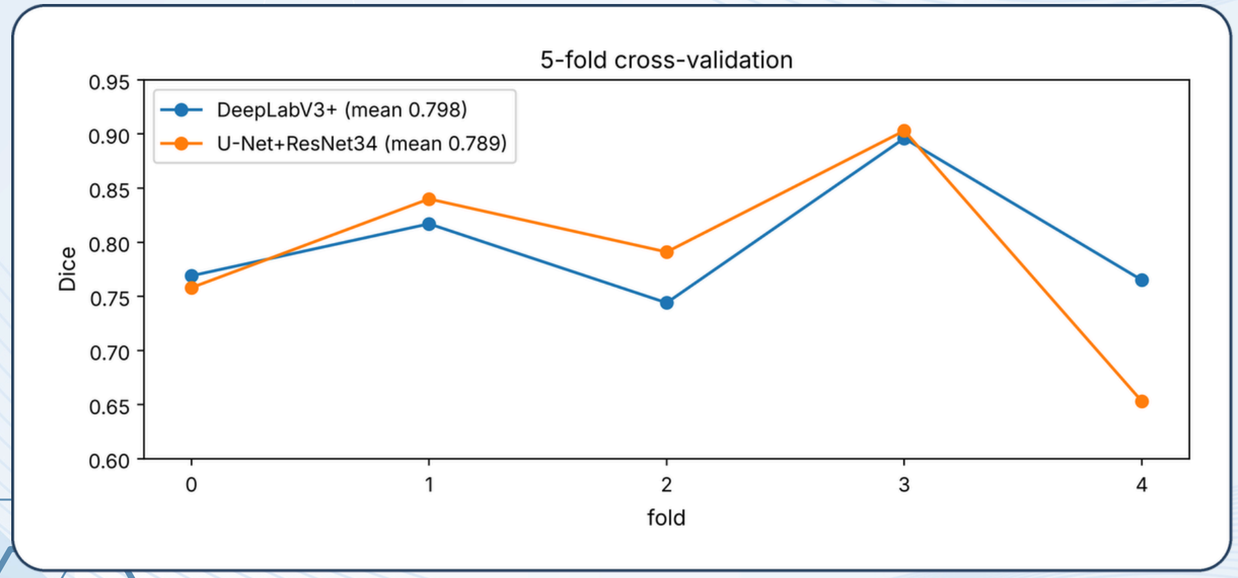
61.8% IoU

93.0% PRECISION

KEY FINDINGS

- Best Dice and IoU
- Strong positive prediction reliability
- Transfer Learning improves the U-Net baseline
- Autoencoder produces many false positives

CLASSIFICATION CROSS-VALIDATION



MEAN DICE COMPARISON

Architecture	Mean Dice	Std
U-Net + ResNet34	0.789	0.094
DeepLabV3+	**0.798**	**0.061**

DEEPLABV3+

0.798 ± 0.061

mean Dice ± std

U-NET + RESNET34

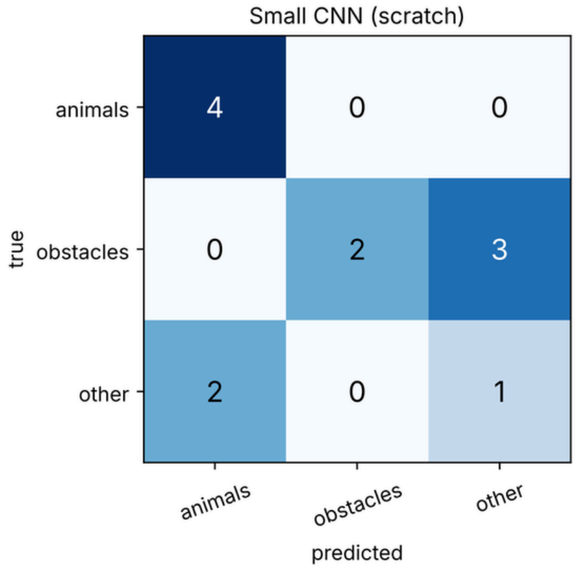
0.789 ± 0.094

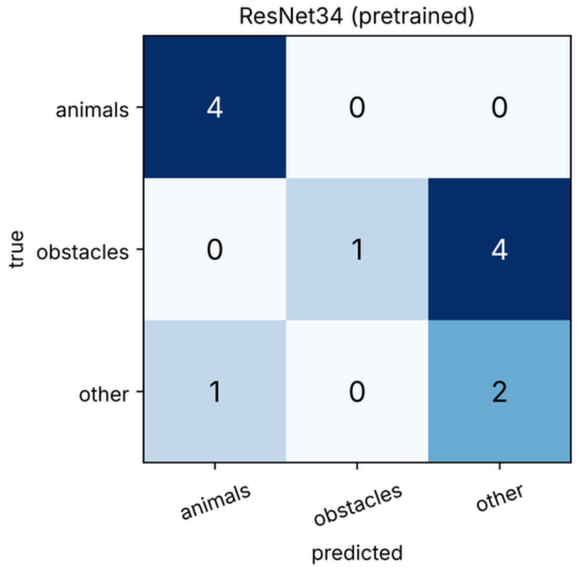
mean Dice ± std

KEY FINDINGS

- DeepLabV3+ has the highest average Dice.
- Only 0.009 difference between the means.
- Lower variability for DeepLabV3+.

CLASSIFICATION RESULTS





FIVE-FOLD CROSS-VALIDATION

Metric	Small CNN	ResNet34
Accuracy	0.435	**0.784**
Macro-F1	0.381	**0.701**
Accuracy Std	0.081	**0.124**
Macro-F1 Std	0.099	**0.135**

KEY FINDING

ResNet34 achieves substantially better average classification performance across the five folds.

INTERPRETATION

WHAT DO THE RESULTS TELL US?

ZONE 2 · VISUAL SEGMENTATION

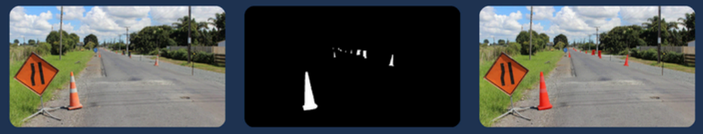
ORIGINAL IMAGE > GROUND TRUTH > OVERLAY

1. Animal in a mountain road

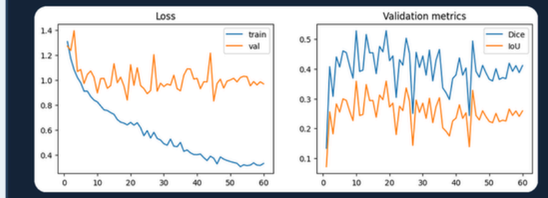
2. Wild animal on a road

3. Multiple animals

4. Road traffic cones



ZONE 1 · TRAINING BEHAVIOR



- Training loss decreases consistently.
- Validation loss fluctuates.
- Validation metrics remain variable.

Interpretation: Possible overfitting and unstable generalization.

ZONE 3 · MODEL COMPARISON

OBSERVATION	INTERPRETATION
DeepLabV3+ leads test Dice	Strongest test segmentation result
ResNet34 leads classification CV	Transfer Learning is beneficial
Autoencoder has low precision	Too many false alarms
Large fold variability	Limited dataset stability

MAIN TAKEAWAY

MODEL PERFORMANCE DEPENDS ON THE TASK, DATASET SIZE AND GENERALIZATION ABILITY.

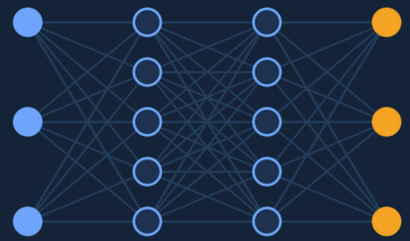
Application Demonstration

FROM DEEP LEARNING MODELS TO A WORKING APPLICATION

1 STEP 1
INPUT IMAGE

Road scene

2 STEP 2
AI INFERENCE



Segmentation | Classification

Trained deep learning models

3 STEP 3
RESULTS

CATEGORY
Animals

LOCATION
Pixel-level mask

RELIABILITY
Model confidence

Overlay of the detected anomaly

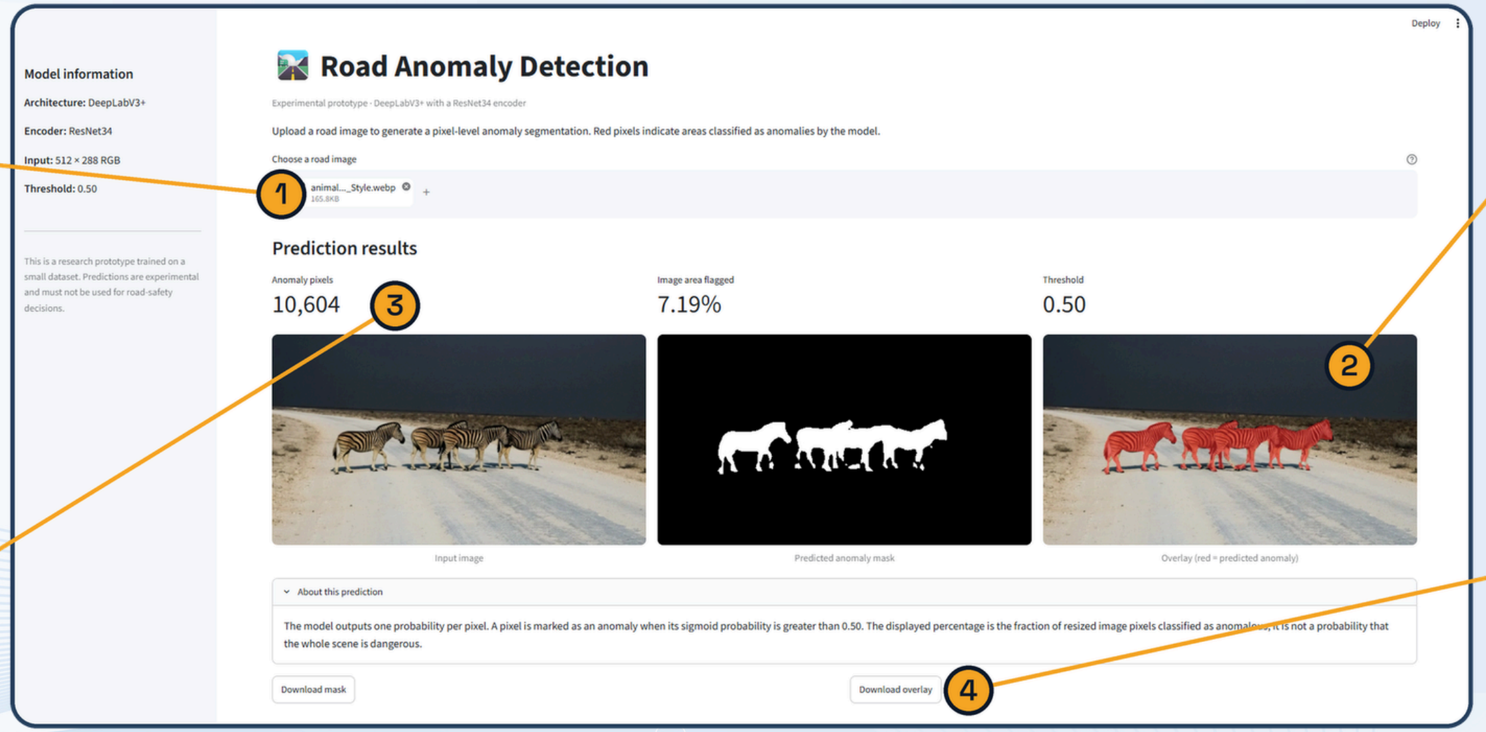
APPLICATION OVERVIEW

1 Upload a road image

Choose any road photo to analyse.

3 Instant prediction metrics

Anomaly pixels, flagged image area and threshold (0.50).



2 Pixel-level segmentation

DeepLabV3+ with a ResNet34 encoder marks anomalous pixels in red.

4 Download results

Export the predicted mask and the red overlay.

CONCLUSIONS and NEXT STEPS

CONCLUSIONS and NEXT STEPS

OUR ACHIEVEMENTS

0.764
Best test Dice

0.618
Best test IoU

0.798
Best cross-validation mean Dice

78.4%
Best classification CV accuracy

FUTURE ROADMAP

1 STEP 1 — DATA

- Larger and more diverse dataset
- Class balancing and improved annotations

→

2 STEP 2 — MODELS

- Explore YOLO and Transformer architectures
- Hyperparameter optimization

→

3 STEP 3 — DEPLOYMENT

- Real-time video processing
- Optimized inference
- Intelligent driving applications

FINAL MESSAGE

“Recognizing an anomaly is only the first step. Understanding its category, location and reliability is what makes road perception intelligent.”

Thank You !